

BRILLIANT EDUCATIONS PUBLISHERS

Kenya Certificate of Secondary Education— Prediction 1

231/1

BIOLOGY

Paper 1



May 2020 - 2 hours



Name Index Number

Signature Date

Instructions to Candidates

- Write your name and index number in the spaces provided above.
- Sign and write date of examination in the spaces provided above.
- Answer **all** the questions in this question paper.
- All answers must be written in the spaces provided.
- This paper consists of 10 printed pages.
- The candidates should check the question paper to ascertain that all pages are printed as indicated and no questions are missing.
- Candidates should answer the questions in English.

For Examiner's Use Only

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21

22	23	24	25	26	27	28	29	30	31	32

Grand Total



© The Brilliant Educations Publishers Examinations
231/1

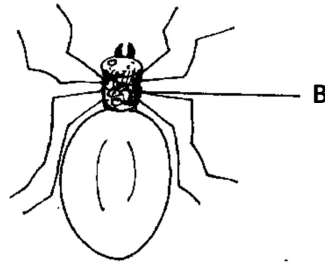


1. Name the bond that exists between amino acids during condensation process of forming proteins? (1mark)

.....

.....

2. A student discovered the organism below moving about on the teacher's desk



- (a) Name the class of which the organism belongs. (1mark)

.....

.....

- (b) Name the part labelled B. (1mark)

.....

.....

3. A certain plant has the following characteristics: presence of roots, stems and the leaves, life cycle in saprophyte and gametophyte generation. Saprophyte generation being the dominant type. Name the division to which the plant belongs. (1mark)

.....

.....

4. State the biological significance of each of the following.

- (a) Thick muscular walls and narrow lumen in arteries. (1mark)

.....

.....

- (b) Narrow xylem vessels in flowering plants. (1mark)

.....

.....

5. What are **two** source of genetic diversity in meiosis ? (2marks)

.....

.....

.....

6. Explain why primary productivity decreases with depth in aquatic environment. (2marks)

.....

.....

.....

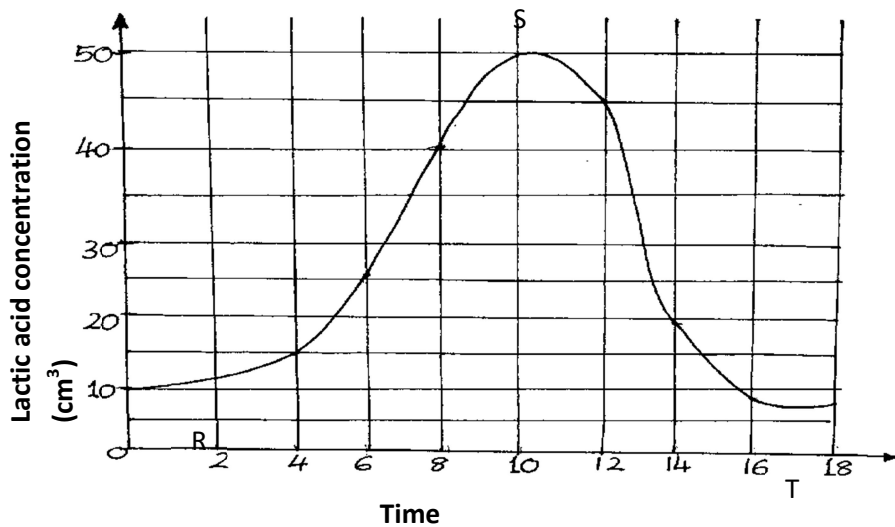
7. Name **two** regions in higher plants where cells actively undergo mitosis. (2marks)

.....

.....

.....

8. The diagram below shows the general appearance of lactic acid in the blood of an athlete after an exercise. Study it carefully and answer the questions that follow.

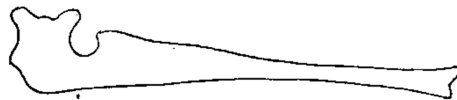


- (a) Name the physiological process represented by the above diagram (1mark)

- (b) Explain what was happening in the body between points: (2marks)

9. Name the class in the phylum arthropoda which has the larger number of individuals. (1mark)

10. A bone obtained from a mammal is represented by the diagram below.



- (a) Name the bone (1mark)

- (b) Which bones articulates with the bone shown in the diagram at the notch? (2marks)

11. (a) (i) What is meant by vestigial structures? (1mark)

.....

.....

(ii) Give an example of a vestigial structure in any other organism other than human? (1mark)

.....

.....

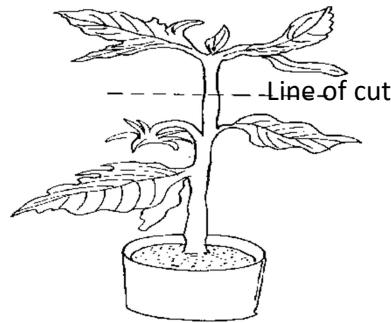
(c) Explain why certain drugs become infective in curing a disease after many years of use (2marks)

.....

.....

.....

12. In an experiment the shoot tip of a young tomato plant decapitated as shown in the diagram below.



(a) State the expected results after 2 weeks (1mark)

.....

(b) Give a reason for your answer in (a) above (2marks)

.....

.....

13. State **three** structural differences between Deoxynbonocleic acid (DNA) and Ribonucleic acid (RNA)? (3 marks)

.....

.....

.....

14. Explain why plants do not require specialized excretory organs. (3marks)

.....

.....

.....

.....

15. Explain how the following factors affect the rate of photosynthesis

(a) Concentration of carbon (iv) oxide. (1mark)

.....

.....

.....

 (b) Intensity

(1mark)

-

 16. Name **three** distinguishing characteristics that make man more developed and superior to all other mammals. (3marks)

-

 17. State **two** advantages of cross-pollination. (2marks)

-

 18. State the role of the following parts in the mammalian intestine

(a) Lacteal in the villi

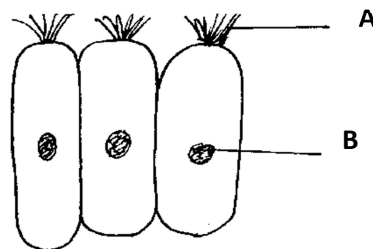
(1mark)

.....

 (b) Goblet cells

(1mark)

-
 19. Study the figure below which shows a type of epithelial tissue



(a) State the name of structure A. (1mark)

.....

 (b) Give an example in humans where this epithelium is found (1mark)

(1mark)

20. A mother of blood group **AB** was married to a man of blood group **B**. What is the probability that one of their children would be blood group **A**. Show your workings. (3marks)

21. Name the substance in plasma and red blood cells that transport carbon (iv) oxide. (2marks)

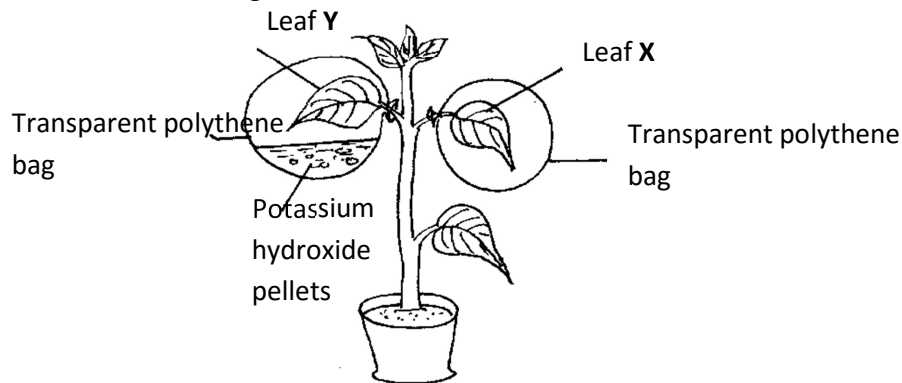
(i) Plasma

.....
.....

(ii) Red blood cells

.....
.....

22. A potted plant was kept in dark for 48 hours. Two leaves **X** and **Y** were treated as shown in the diagram below.



The experimental set up was kept in the sunlight for 6 hours after which a starch test was carried out on the two leaves.

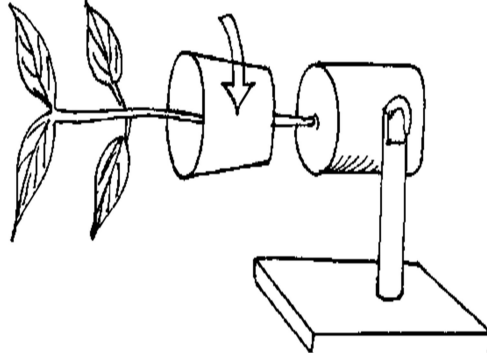
- (a) What were the results of the starch test on leaves **X** and **Y**? (2marks)

.....
.....
.....

- (b) Give reasons for your answers in (a) above (2marks)

.....

23. Carefully study the figure which rotates making one revolution in 15 minutes. A seedling with a straight radicle and plumule was attached to the apparatus as shown below.



- (a) What is the name of the apparatus shown (1 mark)

.....

- (c) Make a drawing of how the seedling might have appeared after one week. (1 mark)

24. (a) What is meant by the term non-disjunction? (1 mark)

.....

- (b) Give an example of genetic disorders caused by:

- (i) Non-disjunction (1 mark)

.....

- (ii) Gene mutation

.....

25. To estimate the population size of crabs in a certain lagoon, traps were laid at random, 400 crabs were caught, marked and released back into the lagoon. Four days later traps were laid again and 374 crabs were caught. Out of the 374 crabs, 80 were found to have been marked.

(a) Calculate the population size of the crab in the lagoon using the formula below:

$$N = \frac{n \times M}{m}$$

Where:

N = Total population of crabs in the lagoon

n = Total number of crabs in the second catch

M = Number of marked crabs during the first catch

m = Number of marked crabs in the second catch. (2marks)

(b) State the assumptions that were made during the observation (2marks)

.....

(d) What is the name given to this method of estimating the population size (1mark)

.....

26. State the role of light in the process of photosynthesis. (1mark)

.....

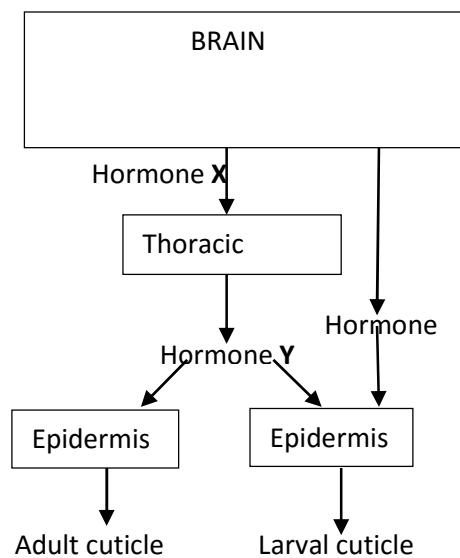
27. When are the **two** organisms considered to belong to the same species. (1mark)

.....

28. State the role of insulin in the human body (1mark)

.....

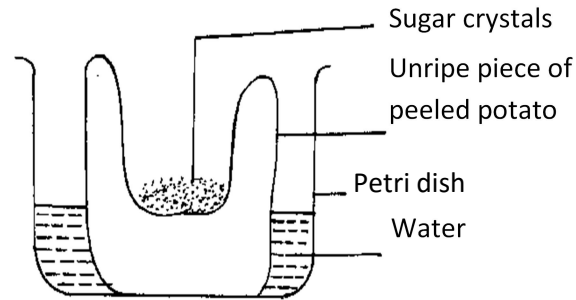
29. The chart below is a summary of the role of hormones in insect metamorphosis



- (a) Name each of the hormones represented by letters **X** and **Y**. (2marks)
-
-
- (b) State the function of hormone **Z** (1mark)
-
-
- (c) What is the advantage of production of hormone **Y** alone without the secretion of hormone **Z**. (1mark)
-
-
30. Give **two** ways in which the mammalian uterus becomes adapted to implantation of zygote (2marks)
-
-
-

31. A form one student trying to estimate the size of onion cells observes the following on the microscope's field of view. He counts 20 cells across the field of view which is 3mm. calculate the size of one cell in micrometers. (3marks)

32. An experiment to investigate a certain physiological process was set up as shown below. After three (3) hours it was observed that the level of the sugar solutions had risen.



- (a) Name the physiological process being investigated? (1mark)

.....

.....

- (b) Account for the rise of the level of sugar solution in the experiment (3marks)

.....

.....

.....



THIS IS THE LAST PRINTED PAGE.

© Kenya Certificate of Secondary – Prediction 1, 2020

0711410583

231/2